

Based on the job listing for the **Data Scientist - Emerging Markets** position at the **Commodity Futures Trading Commission (CFTC)** (Announcement: CFTC-DOD-2026-0003), the agency is looking for a senior-level data scientist (CT-13, salary range \$152,377 - \$223,819).

Indeed

To be highly competitive for this specific role, a candidate's resume must explicitly demonstrate:

1. **Advanced Analytics & Financial Surveillance:** Experience conducting exploratory data analysis on large structured/unstructured datasets, predicting anomalies, and building predictive models for risk scoring or market monitoring.

Indeed

2. **End-to-End Automated Pipelines:** Expertise designing, implementing, and troubleshooting automated data science workflows and machine learning pipelines.

Indeed

3. **Advanced Modeling:** Strong foundations in advanced statistical modeling and machine learning using robust environments, with clean deployment practices.

Indeed

4. **Leadership & Strategy:** Experience providing technical guidance, mentoring staff, supporting enterprise data strategy (AI/ML deployment), and presenting complex findings to senior management or executive stakeholders.

Indeed

Below are three tailored resumes of top candidates engineered to perfectly match the CFTC's key selection criteria.

Candidate 1: The FinTech & Market Surveillance Specialist

This candidate focuses heavily on financial market monitoring, predictive risk scoring, and enterprise-level machine learning pipelines.

Russ Conte Chicago, IL | rconte.datascience@email.com | [linkedin.com/in/russ-conte-ds](https://www.linkedin.com/in/russ-conte-ds)

PROFESSIONAL SUMMARY Accomplished Data Scientist with extensive experience building advanced statistical models, machine learning architectures, and automated data

pipelines to evaluate complex markets. Proven track record in conducting deep exploratory data analysis on massive structured and unstructured datasets to isolate trends, relationships, and anomalies. Adept at presenting complex technical findings to senior leadership and advising on enterprise data strategy to drive AI/ML innovation.

Indeed

CORE COMPETENCIES * Modeling & Analytics: Advanced Statistical Modeling, Machine Learning Pipelines, Predictive Analytics, Financial Risk Scoring.

- **Technical Environment:** R (caret, ggplot2, MASS), Python, SQL, Cloud Analytics Platforms.
- **Domain Expertise:** Automated End-to-End Pipelines, Market Surveillance, Competitive Analysis, Large-scale Data Wrangling.
- **Leadership:** Staff Mentorship, Strategic Data Advisory, Senior Stakeholder Presentations.

Indeed

PROFESSIONAL EXPERIENCE Senior Data Scientist & Analytics Lead | Financial Market Analytics Group | Chicago, IL

May 2022 – Present * Design and implement end-to-end automated machine learning pipelines for high-throughput numeric and logistic datasets, significantly improving predictive accuracy for market trends.

- Lead complex exploratory data analysis on massive, highly volatile datasets encompassing thousands of entries, identifying critical patterns, anomalies, and competitive trends across regional markets.
- Develop and deploy robust predictive models and risk-scoring algorithms utilizing foundational statistical frameworks, streamlining data-driven decision-making for executive teams.

Indeed

- Provide technical guidance and mentorship to junior data analysts, establishing best practices for code architecture, validation routines, and reproducible reporting.

Indeed

- Present detailed, high-impact data visualizations and dashboard solutions to C-suite executives, translating complex computational findings into actionable corporate strategy.

Data Scientist / Quantitative Analyst | Apex Staffing & Market Intelligence | Chicago, IL

August 2019 – May 2022 * Conducted deep-dive competitive analyses and market-share modeling for the temporary staffing industry across the Greater Chicago market, synthesizing disparate structural datasets into cohesive growth forecasts.

- Built automated exploratory data analysis (EDA) scripts that consolidated complex multi-chart outputs (including automated layout configurations for boxplots and histograms) to maximize data clarity for strategic reviews.
- Engineered clean data partition routines and model-tuning pipelines using robust validation workflows, avoiding erratic training errors by establishing precise parametric criteria.
- Maintained code repositories, optimized database queries, and collaborated with infrastructure teams to transition standalone data scripts into enterprise-level workflows.

EDUCATION & CERTIFICATIONS * **Master of Science in Data Science / Applied Statistics** * **Bachelor of Science in Mathematics** * *Professional Portfolio*: Active builder of custom open-source analytical packages designed for CRAN standards.

Candidate 2: The Data Infrastructure & Automated Workflows Architect

This candidate excels at the infrastructure, pipeline automation, and computational analytics technology evaluation side of the CFTC requirements.

Sarah Jenkins Washington, DC | sjenkins.data@email.com | [linkedin.com/in/sjenkins-data](https://www.linkedin.com/in/sjenkins-data)

PROFESSIONAL SUMMARY Senior Data Scientist and Systems Architect specializing in designing automated data science workflows, robust data infrastructure, and advanced predictive analytics. Expert in cloud-based analytics platforms and big data technologies, with a strong focus on data governance, data integrity, and compliance auditing. Experienced technical advisor to federal and corporate leaders on AI/ML deployment strategies.

CORE COMPETENCIES * **Systems & Infrastructure**: Automated Workflow Orchestration, Cloud Data Platforms (AWS/Azure), Big Data Technologies (Spark, Hadoop).

- **Data Governance**: Data Integrity Auditing, Compliance Tracking, Strategy Formulation. **Indeed**
- **Advanced Modeling**: Time-Series Forecasting, Unstructured Data Mining, Supervised Learning.

- **Soft Skills:** Technical Mentorship, Cross-functional Collaboration, Strategic Systems Development.

PROFESSIONAL EXPERIENCE Lead Data Science Infrastructure Engineer | Capital Oversight Solutions | Washington, DC

Jan 2023 – Present * Formulate and execute enterprise data strategy regarding the exploration, deployment, and governance of AI/ML models across traditional and digital market segments.

- Design and manage automated data pipelines that ingest, clean, and process billions of rows of structured and unstructured transaction data daily, ensuring 99.9% data integrity.
- Evaluate emerging computational technologies and software platforms to optimize the firm's data infrastructure and analytics capabilities.

Indeed

- Provide strategic advisory to senior management regarding data science governance, algorithmic bias risk, and systems compliance.

Indeed

- Mentor a team of five data engineers and junior scientists on cloud-native model deployment, version control, and containerization.

Senior Data Engineer / Scientist | Federal Financial Systems Corp | Arlington, VA

Mar 2020 – Jan 2023 * Built advanced predictive analytics models to identify systemic risks and anomalous behaviors within institutional financial reports.

- Developed automated data mining routines that parsed unstructured regulatory filings, turning text metrics into structured features for risk scoring.
- Co-authored the organizational data governance playbook, standardizing metadata tracking, validation loops, and pipeline error handling.
- Presented monthly technical capability briefs and discovery reports to internal oversight committees.

EDUCATION * Master of Science in Computer Science (Concentration in Data Engineering)

- **Bachelor of Science in Data Science** ---

Candidate 3: The Unstructured Data & AI/ML Specialist

This candidate brings heavy expertise in processing unstructured text and tabular data to build advanced risk-scoring measures for investigations and surveillance.

David Vance, PhD Alexandria, VA | dvance.phd@email.com | [linkedin.com/in/dvance-ml](https://www.linkedin.com/in/dvance-ml)

PROFESSIONAL SUMMARY Principal Machine Learning Scientist with a PhD in Computational Statistics and 7+ years of experience leveraging AI and ML to solve complex business and investigative challenges. Specialized in combining structured tabular data with unstructured text analytics to build high-accuracy risk scoring models. Proven capability in providing technical direction for large-scale data exploration initiatives and delivering data-driven recommendations to high-level policy makers.

CORE COMPETENCIES * AI/ML Expertise: Natural Language Processing (NLP), Tabular Risk Scoring, Deep Learning, Ensemble Methods.

- **Computational Skills:** Python, R, Cloud Analytics, Advanced Research Methodologies.
- **Strategic Impact:** Policy-driven Model Generation, Investigation Target Identification, Complex Trend Discovery.

Indeed

- **Leadership:** Academic and Professional Mentorship, Technical Project Management.

PROFESSIONAL EXPERIENCE **Principal Machine Learning Scientist** | Vanguard Risk Intelligence | McLean, VA

Nov 2022 – Present * Develop cutting-edge AI/ML models for multi-tiered risk scoring, successfully identifying high-probability candidates for regulatory and fraud investigations.

- Champion the integration of unstructured data streams (e.g., communications, legal filings, web scrapes) with traditional tabular market data to enhance predictive monitoring.
- Provide technical guidance to multi-disciplinary teams of analysts, investigators, and developers on complex computational research capabilities.

Indeed

- Advise executive leadership on the strategic deployment of enterprise analytics to achieve policy and compliance goals.

Indeed

- Design rigorous testing frameworks to benchmark model performance, stability, and reproducibility before production deployment.

Senior Data Scientist | National Analytics & Security Partners | Washington, DC

Sep 2019 – Nov 2022 * Conducted advanced exploratory data analysis on multi-source datasets to identify novel trends, relationships, and hidden anomalies.

- Utilized advanced statistical methods to build predictive models that strengthened institutional surveillance capabilities.
- Created automated visualization dashboards that translated complex, multi-dimensional statistical models into intuitive decision-making tools for non-technical senior management.
- Mentored graduate-level interns and junior staff on advanced modeling methodologies and experimental design.

EDUCATION * **PhD in Computational Statistics / Mathematics** * **Bachelor of Science in Statistics** ---